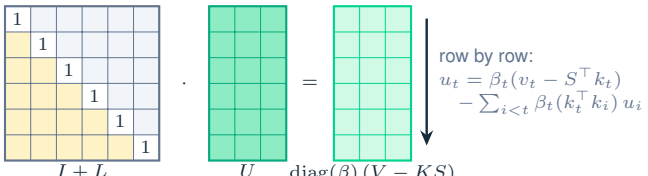
1. Solve for the pseudo-values of the chunk

one $C \times C$ system, unit lower triangular, solved row by row

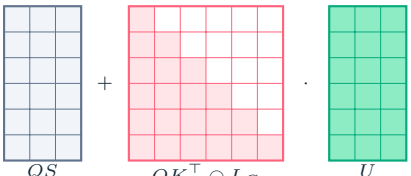


$L_{ti} = \beta_t\, k_t^\top k_i$ for $i < t$

the right side: β_t times each new value minus what S reads

2. Then it is chunked linear attention, with U in place of V

the masked product inside the chunk, the state across chunks



$$O = QS + (QK^\top \odot L_C)\, U$$

$$S \leftarrow S + K^\top U$$

Q, K, V, β: the rows of the chunk

L_C: the lower-triangular $C \times C$ matrix of ones

QS reads all earlier chunks; the masked product reads the pairs inside this chunk

Why the chunk is exact. The chunk multiplies the state by its C transitions, and their product has a compact form:

$$(I - \beta_C k_C k_C^\top) \cdots (I - \beta_1 k_1 k_1^\top) = I - K^\top T K, \qquad T = (I + L)^{-1}\,\mathrm{diag}(\beta), \qquad U = T\,(V - KS).$$

This is the compact WY representation of a product of generalized Householder matrices (Schreiber and Van Loan, 1989), with T from the UT transform (Joffrain et al., 2006). No state inside the chunk is ever formed. Yang et al. (2024b) use this form to train DeltaNet in parallel; their equation (10) gives T.